

Understanding Neural Networks beyond Kernel Methods

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Outline

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 - Neural Tangent Kernel
 - Random Feature Model
- 3 Understanding Neural Networks beyond Kernel Methods
 - Approximation Theory
 - Feature Learning
 - Mean-field Analysis
 - Implicit Regularization
- 4 Conclusion

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Background

Neural networks (NN) can perform surprisingly well in practice [Zhang et al., 2016].

- Overparameterized regime, i.e., number of neurons $N >$ number of samples n .
- Despite the nonconvex and nonsmooth landscape, they can fit all the data, achieve zero training error.
- Generalize well on unseen data.

Question

Can we explain it theoretically?

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Neural Tangent Kernel (NTK)

- Neural networks:

$$f_N(x; \theta) = \frac{1}{\sqrt{N}} \sum_{i=1}^N \sigma(\langle w_i, x \rangle) : w_i \in \mathbb{R}^d.$$

- Condition: large initialization, small learning rate, overparameterization ($N = \Omega(\text{poly}(n))$).
- Linearization near initialization:

$$f_N(x; \theta) = f_N(x; \theta_0) + \langle \theta - \theta_0, \nabla f_N(x; \theta_0) \rangle + o(\|\theta - \theta_0\|_2^2).$$

- Optimizing in RKHS with NTK $K(x_i, x_j) = \mathbb{E}_{w_0} \langle \nabla \sigma(w_0, x_i), \nabla \sigma(w_0, x_j) \rangle$, linear convergence rate can be established [Jacot et al., 2018, Du et al., 2019, Arora et al., 2019]...

Random Feature Model (RFM)

- Let $\phi(x; w)$ be a feature function w.r.t. random variable w , ρ_0 is a distribution for w .
- The random feature model is given by

$$f_N(x; \theta) = \frac{1}{N} \sum_{i=1}^N a_i \phi(x; w_i).$$

- Consider RKHS $\mathcal{H}_k$ with $k(x, x') = \mathbb{E}_{w \sim \rho_0} [\phi(x; w) \phi(x'; w)]$.
- $f \in \mathcal{H}_k$ iff there exists $a(\cdot) \in L^2(\rho_0)$ s.t.

$$f(x) = \int a(w) \phi(x; w) d\rho_0(w), \quad \|f\|_{\mathcal{H}_k}^2 = \int a^2(w) d\rho_0(w).$$

- Functions in $\mathcal{H}_k$ can be well approximated (direct and inverse) by RFM in $L^2(\rho_0)$.

Neural Networks in Random Feature Regime

$$f_N(x; \theta) = \sum_{i=1}^N a_i \sigma(\langle w_i, x \rangle).$$

- We draw w_i randomly and then train a_i only.
- Approximate RKHS with $K(x_i, x_j) = \mathbb{E}_{w_0} \sigma(w_0, x_i) \sigma(w_0, x_j)$.
- If we can learn with random features, then we can learn with neural networks, e.g. learn polynomials [Yehudai and Shamir, 2019].

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NNs outperform Kernel Methods

- The above analysis shows that neural networks are **not worse than** kernel methods.
- Neural networks can be much better than kernel methods [Chizat et al., 2018, Arora et al., 2019].

Question

Can we understand the success of neural networks beyond kernel methods?

We introduce several methods:

- Approximation theory
- Feature learning
- Mean-field analysis
- Implicit regularization

Curse of Dimensionality (CoD)

- Inputs in practical problems usually have high dimension: the input dimension of a CIFAR-10 image is $d = 32 \times 32 \times 3 = 3072$, much higher in ImageNet...
- Classical approximation theory suffer from the curse of dimensionality: As d grows, computational cost grows exponentially fast!
- The error $\sim n^{-\alpha/d}$ for n points.
- Neural networks can break CoD.

Barron's theorem

Now we establish approximation theory with an error rate independent of dimension.

Theorem 1 (Barron, 1993)

Suppose $\|x\|_2 \leq r$, $f(x) = \int e^{i\langle x, w \rangle} F(w) dw$, then

$$\text{Approximation error} \leq \frac{g(f)^2}{N}, \quad g(f) = 2r \int \|w\|_2 |F(w)| dw.$$

The approximation error only depends on the regularity of the function f .

The Barron Space

E et al., 2019

Definition 2 (Barron space)

Consider function $f : X \mapsto \mathbb{R}$. Define the “Barron norm”

$$\|f\|_B := \inf_{\rho \in \Psi_f} \mathbb{E}_\rho |a| \|w\|_1.$$

where $\Psi_f = \{\rho : f(x) = \mathbb{E}_\rho a \sigma(w^T x)\}$, the set of possible representations for f . Define the set of Barron functions

$$\mathcal{B} = \{f \in C(X) : \|f\|_B < \infty\}$$

Definition 3 (Path norm)

If $f_N(\mathbf{x}; \theta) = \frac{1}{N} \sum_{j=1}^N a_j \sigma(\mathbf{w}_j^T \mathbf{x})$, then its path norm is defined by

$$\|\theta\|_P = \frac{1}{m} \sum_{k=1}^m |a_k| \|w_k\|_1.$$

Approximation Theory for Two-layer Neural Networks

Theorem 4

For any $f^* \in \mathcal{B}$, there exists a two-layer network $f_m(\cdot; \theta)$ such that

$$\|f^* - f_m(\cdot; \theta)\|_{L^2(\mu)} \lesssim \frac{\|f^*\|_{\mathcal{B}}}{\sqrt{N}}, \quad \|\theta\|_P \lesssim \|f^*\|_{\mathcal{B}}.$$

Theorem 5

Let f^* be a continuous function. Assume there exist a constant C and a sequence of two-layer neural networks $\{f_m\}$ such that

$$\frac{1}{N} \sum_{k=1}^N |a_k| \|w_k\|_1 \leq C, \quad m \in \mathbb{N}^+, \quad f_m(x) \rightarrow f^*(x)$$

for all $x \in X$, then $f^* \in \mathcal{B}$, i.e. there exists a probability distribution ρ^* on Ω , such that

$$f^*(x) = \int_{\Omega} a \sigma(w^T x) \rho^*(da, dw) = \mathbb{E}_{\rho^*} [a \sigma(w^T x)]$$

for all $x \in X$ and $\|f^*\|_{\mathcal{B}} \leq C$.

Compare NN, NT, RF

We compare the approximation power of three function classes.

Neural Network

$$\mathcal{F}_{NN}^N := \{f_{NN}(x; b, W) = \sum_{i=1}^N b_i \sigma(\langle w_i, x \rangle) : b_i \in \mathbb{R}, w_i \in \mathbb{R}^d, i \in [N]\}.$$

Neural Tangent

$$\mathcal{F}_{NT}^N := \{f_{NT}(x; S, W) = \sum_{i=1}^N \langle s_i, x \rangle \sigma'(\langle w_i, x \rangle) : s_i \in \mathbb{R}^d, i \in [N]\}.$$

Random Feature

$$\mathcal{F}_{RF}^N := \{f_{RF}(x; a, W) = \sum_{i=1}^N a_i \sigma(\langle w_i, x \rangle) : a_i \in \mathbb{R}, i \in [N]\}.$$

Approximation error:

$$R(f_x, \mathcal{F}) = \inf_{f \in \mathcal{F}} \|f_x - f\|_{L^2}^2.$$

Neural Networks outperform Kernel Methods

- Spiked covariates model: $x = (x_1, x_2) \in \mathbb{R}^d$, $x_1 \in \mathbb{R}^{d_s}$, $x_2 \in \mathbb{R}^{d-d_s}$, $d_s = d^\eta$, $0 \leq \eta \leq 1$, $\text{Cov}(x_1) = \text{snrf} \cdot I_{d_s}$, $\text{Cov}(x_2) = I_{d-d_s}$, $\text{snrf} = d^k$, $0 \leq k < \infty$ (feature SNR).
- Response depend on signal features: $y = f_x(x) + \epsilon$, $f_x(x) = \varphi(x_1)$.
- Effective dimension: $d_{\text{eff}} = d_s \vee (d/\text{snrf})$. We have $d_s \leq d_{\text{eff}} \leq d$.

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- Response depend on signal features: $y = f_x(x) + \epsilon$, $f_x(x) = \varphi(x_1)$.
- Effective dimension: $d_{\text{eff}} = d_s \vee (d/\text{snrf})$. We have $d_s \leq d_{\text{eff}} \leq d$.

Theorem 6 (Ghorbani et al., 2020)

Assume $d_{\text{eff}} t^{t+\delta} \leq N \leq d_{\text{eff}} t^{t+1-\delta}$ and “generic condition” on σ , we have

$$R(f_x, \mathcal{F}_{RF,N}(W)) = \|P_{>e} f_x\|_{L^2}^2 + o_d \mathbb{P}(\cdot),$$

$$R(f_x, \mathcal{F}_{NT,N}(W)) = \|P_{>e+1} f_x\|_{L^2}^2 + o_d \mathbb{P}(\cdot).$$

On the contrary, assume $d_s t^{t+\delta} \leq N \leq d_s t^{t+1-\delta}$, we have

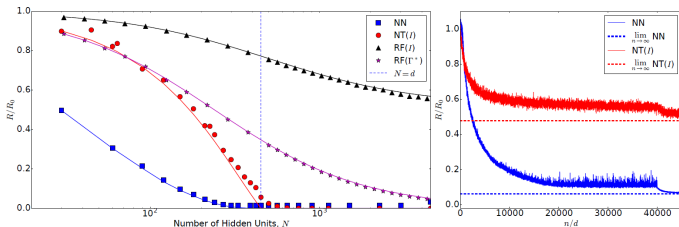
$$R(f_x, \mathcal{F}_{NN,N}) \leq \|P_{>e+1} f_x\|_{L^2}^2 + o_d(1).$$

Moreover, $R(f_x, \mathcal{F}_{NN,N})$ is independent of snrf .

Limitation of Lazy Training

Ghorbani et al., 2019

Data assumption: $x_i \sim \mathcal{N}(0, \mathbf{I}_d)$, $y_i = f_*(x_i) = b_0 + \langle x_i, \mathbf{B}x_i \rangle$, with $\mathbf{B} \succeq 0$. Let $N/d \rightarrow \rho \in (0, 1]$. We compare NT, RF with SGD-trained NN.



$$\frac{R_{M,N}(f_*)}{\|f_*\|_{L_2}^2} \approx \begin{cases} 1 - \frac{\rho}{1+\rho} \frac{\text{Tr}(\mathbf{B})^2}{d\|\mathbf{B}\|_F^2} & \text{for } M = \text{RF}, \\ (1-\rho)_+^2 + \rho(1-\rho)_+ \frac{\text{Tr}(\mathbf{B})^2}{d\|\mathbf{B}\|_F^2} & \text{for } M = \text{NT}, \\ 1 - \frac{\sum_{i=1}^{d \wedge N} \lambda_i(\mathbf{B})^2}{\|\mathbf{B}\|_F^2} & \text{for } M = \text{NN}. \end{cases}$$

Limitations of Random Features

Yehudai and Shamir, 2019

- Gaussian input, hinge loss.
- Random features are hard to learn linear predictors:
Let w^* be a unit vector, r random features $\langle w_i, x \rangle$, $1 \leq i \leq r$. Suppose $r \leq d/2$. Then w.h.p.

$$\left| \sum_{i=1}^r a_i \langle w_i, x \rangle - \langle w^*, x \rangle \right| \geq \frac{1}{4}.$$

- Random features can not be used to learn even a single ReLU neurons, unless the network size is exponentially large in d . However, it is learnable with gradient-methods for neural networks.

Feature Learning

- Neural networks can learn data-dependent features, which can capture the low dimensional and hierarchical structures. While kernel methods use fixed features, leading to worse generalization performance.
- A lot of works train neural networks on toy models using first-order methods. They assume the model have some latent structures and show that NNs gradually learn the features along the training dynamics [Daniely et al., 2020, Bietti et al., 2022, Damian et al., 2022, Mousavi-Hosseini et al., 2023].

Feature Learning

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- The analysis mainly follows the following steps:
 - 1 The learned function $f = g(Ux)$, $U : \mathbb{R}^d \rightarrow \mathbb{R}^r$, Gaussian, boolean data.
 - 2 Gradient of the population loss at initialization can distinguish features.
 - 3 Train the first layer to align with the latent structure.
 - 4 Optimize the second layer until convergence.
 - 5 The number of samples and neurons are polynomial or linear in d , much better than kernel methods.

Learning Parities by Neural Networks

Daniely et al., 2020

- $\mathcal{X} = \{\pm \frac{1}{\sqrt{d}}\}^d$, $\mathcal{Y} = \{\pm 1\}$, hinge loss.
- For $A \subset [d]$, $|A| = k$, let $f_A(x) = \text{sign}(\prod_{i \in A} x_i)$. $\mathcal{D}_A^1$: uniform distribution on $\mathcal{X}$. $\mathcal{D}_A^2$: uniform distribution on $[d] \setminus A$, the bits in A are all 1 w.p. $\frac{1}{2}$ and -1 w.p. $\frac{1}{2}$. Let $\mathcal{D}_A$ be the distribution with $x \sim \mathcal{D}_A^1$ w.p. $\frac{1}{2}$ and $x \sim \mathcal{D}_A^2$ w.p. $\frac{1}{2}$. $y = f_A(x)$.
- We have a family of distributions $\mathcal{F} = \{\mathcal{D}_A : A \subset [d], |A| = k\}$
- For any linear method, there exists some $\mathcal{D}_A \in \mathcal{F}$ that can only be learned with exponential size.
- NNs can learn parities in polynomial size and steps using online SGD.
- After the first gradient step, the gradient is large for bits in the parity and small for bits outside the parity.
- For sufficiently large network and small learning rate, the weights of the first layer moves slow.

Learning Single-index Models

Recall that Hermite polynomials $\{h_j\}_{j \in \mathbb{N}}$ form an orthogonal basis of $L^2(\gamma)$, where γ is the standard Gaussian distribution. It can be used to measure the complexity of the target link function:

Definition 7 (Information exponent, Arous et al., 2021)

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be any function such that $f \in L^2(\gamma)$. The information exponent of f , which we denote by s , is the index of the first non-zero Hermite coefficient.

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- Single-index model:

$$y = f_*(\langle u^*, x \rangle) + \xi,$$

where $u^* \in \mathbb{S}^{d-1}$, $\xi \sim \mathcal{N}(0, \sigma^2)$.

- Assume f_* is known, estimate u^* by θ . Let $m = \langle u, u^* \rangle$.
- $n = O(d^s)$ samples are sufficient to escape the "equator" $\{u : m = 0\}$.

Learning Single-index Models

- When the link function f_* is unknown, we can use random features [Bietti et al., 2022].

-

$$f_N(x; a, u) = a^\top \Phi(\langle u, x \rangle) = \frac{1}{\sqrt{N}} \sum_{i=1}^N a_i \sigma(\varepsilon_i \langle u, x \rangle - b_i),$$

with random features $\Phi(\theta) = \frac{1}{\sqrt{N}}(\sigma(\varepsilon_i \theta - b_i))_{i \in [N]}$, $b_i \sim \mathcal{N}(0, \tau^2)$, ε_i are Rademacher random variables.

- Under some regularity assumptions, f_* is well approximated by RKHS w.r.t. random features $\Phi(\theta)$.
- First-order critical points: either $u \in \{\pm u^*\}$ or $m = 0$.
- Two-stage training of gradient flow finds approximate minimizers.

Learning Multiple-index Models

Damian et al., 2022

- $\mathcal{X} \sim \mathcal{N}(0, \mathbf{I}_d)$, $y_i = f_*(x_i) + \epsilon_i$, $\epsilon_i \in \{-\xi, \xi\}$. Consider multiple-index model:

$$f_*(x) = g(\langle x, u_1 \rangle, \dots, \langle x, u_r \rangle),$$

g is the link function, here we assume it is a $\deg(p)$ - polynomial.

- The eigenspace $S := \text{span}(u_1, \dots, u_r)$. Assume $H := \mathbb{E}_{x \in \mathcal{D}}[\nabla^2 f_*(x)]$ has rank r , i.e. $\text{span}(H) = S$. We use MSE loss.
- Using layerwise training, two layer-neural networks can learn with $n \gtrsim dr^p + d^2r$ samples.
- Kernel methods need $\Omega(d^p)$ samples.
- Change the link function g , neural networks behaves well in transfer learning.
- Key insight: taking Hermite expansion, the coefficient (information exponent) implies the subspace H .

Learning Multiple-index Models

- SGD-trained neural networks can learn low-dimensional representations, which helps to derive better generalization bound [Mousavi-Hosseini et al., 2023].
- Huber loss:

$$\ell_H(\hat{y} - y) := \begin{cases} \frac{1}{2}(\hat{y} - y)^2 & |\hat{y} - y| \leq 1, \\ |\hat{y} - y| - \frac{1}{2} & |\hat{y} - y| > 1. \end{cases}$$

- Gradient of the regularized population loss align with the principal space:

$$\nabla_W L_\lambda(W) = (H(W) + \lambda \mathbf{I})W + D(W)S.$$

- Obtain generalization bound $\sqrt{\frac{dr}{T}}$.
- ReLU NN can learn any monotone polynomial with a sample complexity linear in d .

Two-layer Neural Networks

- Consider neural networks as follows:

$$f_N(x; \theta) = \frac{1}{N} \sum_{i=1}^N \sigma_*(x; \theta_i),$$

where $\sigma_*(x; \theta_i)$ is the activation function, i.e., $\sigma_*(x; \theta_i) = a_i \sigma(\langle x, w_i \rangle + b_i)$.

- It can be represented with distribution $\rho \in \mathcal{P}(\mathbb{R}^d)$:

$$f(x; \rho) = \int \sigma_*(x; \theta) d\rho(\theta)$$

through $\rho_N = \frac{1}{N} \sum_{i=1}^N \delta_{\theta_i}$

Quadratic Population Risk

- We consider square loss and population risk:

$$\begin{aligned} R_N(\theta) &= \mathbb{E}_{\mathbf{x}, y} \left[\left(y - \frac{1}{N} \sum_{i=1}^N \sigma_*(\mathbf{x}; \theta_i) \right)^2 \right] \\ &= \mathbb{E} y^2 + \frac{2}{N} \sum_{i=1}^N V(\theta_i) + \frac{1}{N^2} \sum_{i,j=1}^N U(\theta_i, \theta_j), \end{aligned}$$

where

$$V(\theta_i) = -\mathbb{E}[y \sigma_*(\mathbf{x}; \theta_i)], \quad U(\theta_i, \theta_j) = \mathbb{E}[\sigma_*(\mathbf{x}; \theta_i) \sigma_*(\mathbf{x}; \theta_j)].$$

- Similarly, we can define $R(\rho)$ for $\rho \in \mathcal{P}(\mathbb{R}^d)$,

$$R(\rho) = \mathbb{E} y^2 + 2 \int V(\theta) d\rho(\theta) + \int U(\theta_1, \theta_2) d\rho(\theta_1) d\rho(\theta_2).$$

Here V, U are two potentials.

- Can we understand $R_N(\theta)$ by analyzing $R(\rho)$?

Mean-field Analysis

- Online SGD for θ^k , with $(x_k, y_k) \sim \mathbb{P}_{x,y}, i \in [N]$,

$$\theta_i^{k+1} = \theta_i^k - s_k \nabla_{\theta_i^k} R_N(\theta^k).$$

- Step size $s_k = \varepsilon \xi(k\varepsilon)$ for some regularity function $\xi(\cdot)$,
 $k = t/\varepsilon, N \rightarrow \infty, \varepsilon \rightarrow 0$:

$$\rho_N^k := \frac{1}{N} \sum_{i=1}^N \delta_{\theta_i^k} \Rightarrow \rho_t.$$

- Continuous equation:

$$\partial_t \rho_t(\theta) = -\nabla_{\theta} \cdot (\rho_t(\theta) v(\theta; \rho_t))$$

for some velocity field $v(\theta; \rho_t)$.

- Distributional dynamics for ρ_t [Mei et al., 2018],

$$\partial_t \rho_t(\theta) = 2\xi(t) \nabla_{\theta} \cdot (\rho_t(\theta) \Psi(\theta; \rho_t)),$$

where $\Psi(\theta; \rho) = \frac{\delta R(\rho)}{\delta \rho(\theta)} = V(\theta) + \int U(\theta, \theta') d\rho(\theta')$.

Assumptions

Take $\theta = (a, w)$ with $a \in \mathbb{R}$ and $w \in \mathbb{R}^{d-1}$ and activation $\sigma_*(x; \theta) = a\sigma(x; w)$. Denote $V(\theta) = av(w)$ and $U(\theta_1, \theta_2) = a_1a_2u(w_1, w_2)$ where

$$v(w) = -\mathbb{E}\{\nu\sigma(x; w)\}, \quad u(w_1, w_2) = \mathbb{E}_x\{\sigma(x; w_1)\sigma(x; w_2)\}.$$

Assumptions:

A1. $\sigma : \mathbb{R}^d \times \mathbb{R}^{d-1}$ and ν are bounded, i.e., $\|\sigma\|_\infty, |\nu| \leq K_1$. For any w , $\nabla_w \sigma(x; w)$ is K_1 -sub-Gaussian with respect to $x \sim \mathbb{P}$.

A2. Functions $w \mapsto v(w)$ and $(w_1, w_2) \mapsto u(w_1, w_2)$ are differentiable with bounded and Lipschitz gradients:

$$\|\nabla v(w)\|_2 \leq K_2, \quad \|\nabla u(w_1, w_2)\|_2 \leq K_2,$$

$$\|\nabla v(w) - \nabla v(w')\|_2 \leq K_2\|w - w'\|_2,$$

$$\|\nabla u(w_1, w_2) - \nabla u(w'_1, w'_2)\|_2 \leq K_2\|(w_1, w_2) - (w'_1, w'_2)\|_2.$$

A3. Initialization $\rho_0 \in P(\mathbb{R}^d)$ is supported on $|a_i| \leq K_3$.

Approximation Error

Consider two cases:

General coefficients: initialize parameters $\theta_i^0 = (a_i^0, w_i^0)$ as $(\theta_i^0)_{i \leq N} \sim_{iid} \rho_0$. Update both a_i and w_i during the dynamics.

Fixed coefficients: same initialization but only update w_i during the dynamics.

Theorem 8 (Mei et al., 2019)

Let σ_k verifies assumptions A1-A3. Take $T \geq 1$.

Fixed coefficients: $\exists K$ depending only on K_1 - K_3 such that with proba at least $1 - e^{-z^2}$,

$$\sup_{k \in [0, T/\varepsilon] \cup \mathbb{N}} \left| R_N(\theta^k) - R(\rho_k) \right| \leq K e^{KT} \frac{1}{\sqrt{N}} \left[\sqrt{\log N} + z \right] + K e^{KT} \left[\sqrt{d + \log(N)} + z \right] \sqrt{\varepsilon}.$$

General coefficients: same result with $e^{KT} \rightarrow e^{KT^3}$.

- The approximation is accurate for $T, K = O(1), N \gg 1, \varepsilon \ll 1/d$.
- The bound is dimension-free: N does not directly depend on d .

Classifying Centered Anisotropic Gaussians

$R(\rho)$ is convex w.r.t. ρ , however it could be difficult for ρ_t to converge to a global minima. Here we show a concrete example with convergence.

Data distribution $(y, \mathbf{x})$:

- With probability $1/2$: $y = +1$, $\mathbf{x} \sim \mathcal{N}(0, \Sigma_+)$,
- With probability $1/2$: $y = -1$, $\mathbf{x} \sim \mathcal{N}(0, \Sigma_-)$,
- $\Sigma_{\pm} = \mathbf{U} \text{diag}((1 \pm \Delta)^2 \cdot \text{Id}_{s_0}, \text{Id}_{d-s_0}) \mathbf{U}^\top$.
- $\mathbf{U} \in \mathbb{R}^{d \times d}$ orthogonal matrix.
- $\mathbf{P}_{\mathcal{V}}$: projection on the subspace $\mathcal{V} = \text{span}(\mathbf{U}_{1:s_0})$.

We show the convergence of SGD by analyzing the convergence of ρ_t .

Classifying Centered Anisotropic Gaussians

- $(\mathbf{x}, y) \sim \mathbb{P}$ is invariant under $\mathcal{O}(\mathcal{V}) \times \mathcal{O}(\mathcal{V}^\perp)$.
- Denote $r_1 := \|\mathbf{P}_\mathcal{V} \mathbf{w}\|_2$ and $r_2 := \|(\text{Id} - \mathbf{P}_\mathcal{V}) \mathbf{w}\|_2$. If ρ_0 is spherically symmetric, solution ρ_t of DD remains uniform conditional on r_1, r_2 :

$$\rho_t(\mathbf{w}) = \bar{\rho}_t(r_1, r_2) \times \mu_{s_0} \left(\frac{\mathbf{P}_\mathcal{V} \mathbf{w}}{r_1} \right) \times \mu_{d-s_0} \left(\frac{(\text{Id} - \mathbf{P}_\mathcal{V}) \mathbf{w}}{r_2} \right), \quad \mu_p \equiv \text{Unif}(\mathbb{S}^{p-1}).$$

(global optimum must be of this form by Jensen's inequality).

- PDE on $\rho_t \in \mathcal{P}(\mathbb{R}^d)$ simplifies to PDE on $\bar{\rho}_t \in \mathcal{P}([0, \infty)^2)$:

$$\partial_t \bar{\rho}_t(r_1, r_2) = \nabla \cdot (\bar{\rho}_t(r_1, r_2) \nabla \bar{\Psi}(r_1, r_2; \bar{\rho}_t)).$$

- PDE in 2 dimensions: efficient to solve numerically, can classify stationary points... For some good initialization $\bar{\rho}_0$ and activation σ , and denoting $s_0 = \gamma d$, the PDE converges to global optimum in time $T(\Delta, \gamma, \bar{\rho}_0)$.

Classifying Centered Anisotropic Gaussians

- $\Sigma_{\pm} = \mathbf{U} \text{diag}((1 \pm \Delta)^2 \cdot \text{Id}_{s_0}, \text{Id}_{d-s_0}) \mathbf{U}^{\top}$ with $s_0 = \gamma d$ and $\gamma \in (0, 1)$.
- $\sigma(t) = 0$ for $t \leq 0$, $\sigma(t) = 1$ for $t \geq 1$, and $\sigma(t) = t$ for $0 \leq t \leq 1$ (truncated ReLU).
- $(\mathbf{w}_i^0)_{i \leq N} \stackrel{\text{iid}}{\sim} \bar{\rho}_0 \times \text{Unif}(\mathbb{S}^{d-1})$ with $\bar{\rho}_0$ with bounded density and $R_{d=\infty}(\bar{\rho}_0) < 1$.

Theorem 9

For any $\eta, \Delta, \delta > 0$, there exists

$$d_0 \equiv d_0(\eta, \bar{\rho}_0, \Delta, \gamma), \quad C_0 \equiv C_0(\eta, \bar{\rho}_0, \Delta, \gamma), \quad T \equiv T(\eta, \bar{\rho}_0, \Delta, \gamma),$$

such that for $d \geq d_0$, $N \geq C_0 d$ and $\varepsilon \leq 1/(C_0 d)$, we have with probability at least $1 - \delta$,

$$R_N(\theta^k) \leq \inf_{\rho} R(\rho) + \eta/2 \leq \inf_{\theta \in \mathbb{R}^{N \times d}} R_N(\theta) + \eta,$$

where $k = T/\varepsilon$.

Benefits of mean-field theory

- Exploiting symmetries in the data distribution.
- Allow to focus on studying the PDE.

Noisy SGD and Entropy Regularization

Now we establish the general convergence analysis in noisy SGD.

- Noisy SGD:

$$\theta_i^{k+1} = \theta_i^k + \epsilon(y_k - f_N(x_k; \theta^k)) \nabla_{\theta_i} \sigma_*(x_k; \theta_i^k) + \sqrt{\epsilon/\beta} \cdot g_i^k,$$

where $g_i^k \sim_{iid} N(0, \mathbf{I}_d)$.

- Mean-field description:

$$\partial_t \rho_t = \nabla_{\theta} \cdot (\rho_t \nabla_{\theta} \Psi(\theta; \rho_t)) + \frac{1}{\beta} \Delta_{\theta} \rho_t.$$

- Wasserstein gradient flow for the free energy:
 $F_{\beta}(\rho) = R(\rho) + \frac{1}{\beta} \int \rho(\theta) \log \rho(\theta) d\theta.$
- Convergence $\rho_t \Rightarrow \rho_*^{\beta}$ global minimizer $F_{\beta}(\rho)$. Under some regularity conditions,

$$R(\rho_s^{\beta}) \leq \inf_{\rho} R(\rho) + O\left(\frac{D}{\beta}\right).$$

Bounds with Noisy SGD

Theorem 10

Let σ_* verifies assumptions A1-A3, $T \leq K_4$ and $T \geq 1$. **Fixed coefficients:** $\exists K$ depending only on K_1 - K_4 such that with proba at least $1 - e^{-z^2}$

$$\sup_{k \in [0, T/\varepsilon] \cup \mathbb{N}} \left| R_N(\theta^k) - R(\rho_{k\varepsilon}) \right| \leq K e^{KT} \frac{1}{\sqrt{N}} \left[\sqrt{\log N} + z \right] + K e^{KT} \left[\sqrt{D + \log(N)} + z \right] \sqrt{\varepsilon}.$$

General coefficients: $\exists K$ depending only on K_1 - K_4 such that with proba at least $1 - e^{-z^2}$

$$\begin{aligned} \sup_{k \in [0, T/\varepsilon] \cup \mathbb{N}} \left| R_N(\theta^k) - R(\rho_{k\varepsilon}) \right| &\leq K e^{e^{KT} [\sqrt{\log N} + z^2]} \left[\sqrt{D \log N} + \log^{3/2}(NT) + z^5 \right] / \sqrt{N} \\ &\quad + K e^{e^{KT} [\sqrt{\log N} + z^2]} \left[\sqrt{D \log(NT/\varepsilon)} + \log^{3/2}(N) + z^6 \right] \sqrt{\varepsilon}. \end{aligned}$$

General coefficients: harder to control. The bound is not dimension-free and only allows us to control the approximation error up to $T = o(\log \log N)$ instead of $T = o(\log N)$.

Optimal Transport

Chizat et al., 2018

- The distributional dynamics ρ_t can be regarded as gradient flow in Wasserstein space $\mathcal{P}_2(\Omega)$ of $R(\rho)$.
- If ρ_N^0 converges to some $\rho_0 \in \mathcal{P}_2(\Omega)$, then the gradient flow for $R_N(\theta)$ converges to the unique Wasserstein gradient flow of R starting from ρ_0 (as $N \rightarrow \infty$).
- Homogeneous activation: $\sigma_*(x, rw) = r^2 \sigma_*(x, w)$, including ReLU.
- Assume the support of ρ_0 separates the spheres $r_a \mathbb{S}^{d-1}, r_b \mathbb{S}^{d-1}$, then the Wasserstein gradient flow converges to the global minimizer of $R(\rho)$.

Learnability of SGD

- Mean-field analysis helps to capture the nonlinear dynamics of SGD, which can be used to study the learnability of SGD.
- Two-layer neural networks enjoy the uniform approximation power, what functions can be efficiently learned through SGD?
- Function class: $h : \{\pm 1\}^P \rightarrow \mathbb{R}, p \ll d$.

$$h(z) = \sum_{S \subseteq [P]} \hat{h}(S) \mathcal{X}_S(z),$$

where $\mathcal{X}_S(z) = \prod_{i \in S} z_i$, $\hat{h}(S) := \langle h, \mathcal{X}_S \rangle$ with $\langle f, g \rangle = \mathbb{E}_z[f(z)g(z)]$ for $z \sim \text{Unif}(\{\pm 1\}^P)$. The weights $\hat{h}(S)$ is called the Fourier coefficients of h .

- Data distribution: $f_*(x) = h(z)$

[Abbe, Boix-Adsera, M., '22] With probability at least $1 - 1/M$:

$$\sup_{k \in \{0, \dots, T\}} \left\| \hat{f}_{\text{NN}}(\cdot; \Theta^k) - \hat{f}_{\text{NN}}(\cdot; \bar{\rho}_{k\eta}) \right\|_{L^2} \leq K e^{K(\eta T)^7} \left[\underbrace{\sqrt{\frac{P}{d}}}_{d \rightarrow \infty} + \underbrace{\sqrt{\frac{\log(M)}{M}}}_{M \rightarrow \infty} + \underbrace{\sqrt{d\eta}}_{\eta \rightarrow 0} \right]$$

Merged-staircase property

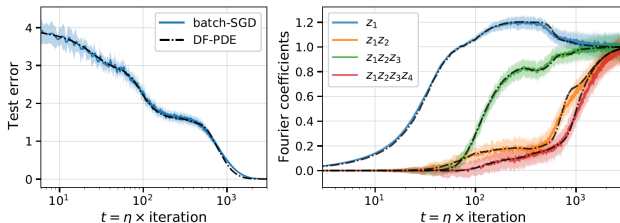
- Leap-1 functions: $z_1 + z_1 z_2 + z_1 z_2 z_3, z_1 + z_2 + z_1 z_2 z_3$.
- Higher leap functions: $z_1 + z_1 z_2 z_3$.

Theorem 11

It is necessary and nearly sufficient for h to be a leap-1 function in order for SGD to converge to 0 test error.

$d = 100, M = 100$:

$$h_*(z) = z_1 + z_1 z_2 + z_1 z_2 z_3 + z_1 z_2 z_3 z_4$$



Mean-field Analysis with Polynomial Size

- The target function: $y(x) = h(q^\top x)$, where $h : \mathbb{R} \rightarrow \mathbb{R}$ is an unknown even quartic polynomial, $q \in \mathbb{R}^d$.
- NN: $f_N(x; \theta) = \frac{1}{N} \sum_{i=1}^N \sigma(\theta_i^\top x)$. σ is a quartic polynomial similar to h .
- Projected gradient flow with sample size $n = O(d^{3.1})$ converges in poly d time. While kernel methods need $\Omega(d^4)$ samples.
- The exponential coupling error is inevitable at the beginning. Compare the growth of coupling error and the growth of the signal. And show that PL condition holds.

Analyzing 1D Dynamics

$$\text{Suppose } \bar{L} = (\mathbb{E}w^2 - \gamma_2)^2 + (\mathbb{E}w^4 - \gamma_4)^2$$

$$\triangleright \frac{\partial L}{\partial w} = 2(\mathbb{E}w^2 - \gamma_2)w + 4(\mathbb{E}w^4 - \gamma_4)w^3$$

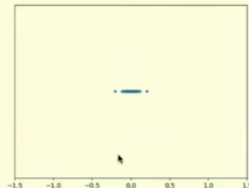
$\triangleright$ Stationary scenarios:

$$\frac{\partial L}{\partial w} = 0, \forall w$$

$\triangleright$ Strategy

- Analyzing all possible stationary points
- Design potential functions to show that w 's are far away from stationary points
- When w 's are far away from stationary scenarios, show

$$\mathbb{E}_w \left(\frac{\partial L}{\partial w} \right)^\alpha \gtrsim \bar{L}$$



Is Exponential Blow-up End of the World?

- ▶ Exponential divergence is unavoidable at the beginning even for 1D dynamics

- recall $w = q^\top u$, and $L \approx (\mathbb{E}w^2 - \gamma_2)^2 + \dots$

- because $\frac{dw}{dt} \approx (\gamma_2 - \mathbb{E}[w^2])w$

- For two init χ_1, χ_2 ,

$$\frac{d(w(\chi_1) - w(\chi_2))}{dt} \approx (\gamma_2 - \mathbb{E}[w^2])(w(\chi_1) - w(\chi_2))$$

- ▶ Rate of exponential divergence is $\approx \gamma_2 - \mathbb{E}[w^2] \approx \gamma_2$
- ▶ At later stage, the divergence slows down because $\gamma_2 - \mathbb{E}[w^2] \approx 0$

- ▶ Idea: compare the error growth with the signal growth

$$\frac{d\|\hat{u}_t - u_t\|_2^2}{dt} \approx (\gamma_2 - \mathbb{E}[w_t^2])\|\hat{u}_t - u_t\|_2^2$$

$$\frac{dw_t^2}{dt} \approx (\gamma_2 - \mathbb{E}[w_t^2])w_t^2$$

Implicit Regularization in Neural Networks

- Neural networks can perform well without explicit regularization. Training neural networks using gradient methods enjoys implicit regularization.
- Analyze the limit of gradient flow on infinitely wide two-layer neural networks. Training both layers and only training the output layer (kernel methods) have different implicit bias. The former can find low-dimensional structures of the data [Chizat et al., 2020].
- NNs perform kernel methods at the beginning (NTK regime)– L_2 regularization.
- Parameters move and learn features (nonlinear dynamics), converge to max-margin solution– L_1 regularization.
- The implicit bias can provably induce grokking [Lyu et al., 2024].

Outline

- 1 Background
- 2 Understanding Neural Networks from Kernel Perspective
 - Neural Tangent Kernel
 - Random Feature Model
- 3 Understanding Neural Networks beyond Kernel Methods
 - Approximation Theory
 - Feature Learning
 - Mean-field Analysis
 - Implicit Regularization
- 4 Conclusion

Conclusion

Summary

- Neural networks can be explained in kernel regimes, including NTK, RFM.
- NNs are more expressive than kernel methods.
- Understanding the nonlinear dynamics of optimization in neural networks through mean-field analysis.
- NNs can adapt to latent structures in data. Useful representations are learned to solve the problems.

Future work

- Understanding neural networks in other regimes.
- Global convergence in mean-field analysis.
- Learning features by standard gradient methods under milder assumptions. Propose a general framework.
- How do neural networks perform in other tasks, i.e., reinforcement learning, LLMs ?